

④ (a)

$$11 = 9m + C$$

$$q = 6m + C$$

$$2 = 3m$$

$$\frac{2}{3} = m$$

$$q = 6\left(\frac{2}{3}\right) + C$$

$$q = 4 + C$$

$$5 = C$$

$$\underline{\underline{m = \frac{2}{3}, C = 5}}$$

(b) $U_4 = \frac{2}{3} \times 11 + 5$

$$= \frac{22}{3} + \frac{15}{3}$$

$$= \frac{37}{3}$$

Question			Generic scheme	Illustrative scheme	Max mark
4.	(a)		•¹ interpret recurrence relation •² interpret recurrence relation •³ find m and c	•¹ $9 = 6m + c$ •² $11 = 9m + c$ •³ $m = \frac{2}{3}$ and $c = 5$	3
Notes:					
1. Correct answer with no working award 0/3. 2. Do not penalise $9 = m6 + c$ or $11 = m9 + c$ at • ¹ and • ² . 3. For candidates who state $m = \frac{2}{3}$, $c = 5$ and then verify that these values work for the given terms, award 2/3.					
Commonly Observed Responses:					
	(b)		• ⁴ calculate term	• ⁴ $\frac{37}{3}$ or $12\frac{1}{3}$	1
Notes:					
4. The answer in (b) must be consistent with the values found in (a). 5. Accept $12.\dot{3}$ or $12.3\dots$ for • ⁴ . Do not accept a rounded answer.					
Commonly Observed Responses:					